

MEA OCPP 1.6 Compliance Test Report

Section 6: Charging Profile Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-11 12:07

Contents

1	Test Configuration	2
2	Section 6 Results	2
3	Notes on Failures and Warnings	3
4	Dependency Map	4

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Plug Timeout	0 s per plug event
Local Stop Timeout	30 s (RFID card tap)
Test Date	2026-06-11 04:41:32 UTC
Results file	tex/vsecc_section6_results.json

Test Architecture

SetChargingProfile commands are issued via the MEA REST API (POST /EV/remote/SetChargingProfile). Two charging sessions are tested:

- 1. Local start with charging profiles (6.1–6.13):** EV plugged in; driver taps RFID card to authorize and start; SetChargingProfile (TxProfile, 5 kW) is applied during charging; a second SetChargingProfile update (7.4 kW) is applied; MeterValues are sampled before and after each profile change; session is terminated by local card tap (or RemoteStop fallback).
- 2. Remote start with charging profiles (6.14–6.26):** EV plugged in; CSMS sends RemoteStartTransaction via MEA REST API; SetChargingProfile (5 kW) is applied, then updated (7.4 kW); MeterValues are sampled after each change; CSMS sends RemoteStopTransaction to end the session.

MQTT subscription to `mqtt://192.168.1.166:1883` is used to observe all StatusNotification, Authorize, StartTransaction, StopTransaction, and MeterValues events.

2 Section 6 Results

Summary: 4 PASS 0 FAIL 22 WARN 0 SKIP (26 total)

Item	Test Description	Detail / Evidence	Result
Flow 1: Local Start with Charging Profile (6.1–6.13)			
6.1	StatusNotification Available (initial state)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
6.2	StatusNotification Preparing (EV plug, Flow 1)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
6.3	Authorize (RFID card) → Accepted	Depends on 6.2 (EV plug)	WARN
6.4	StartTransaction (manual RFID, meterStart, idTag, connectorId)	Depends on 6.2 (EV plug)	WARN
6.5	StatusNotification Charging (Flow 1)	Depends on 6.2 (EV plug)	WARN
6.6	MeterValues (Flow 1, before SetChargingProfile)	Depends on 6.2 (EV plug)	WARN
6.7	SetChargingProfile (TxProfile, 5 kW, stackLevel=1) → Accepted	Depends on 6.2 (EV plug)	WARN
6.8	MeterValues (Flow 1, after first SetChargingProfile)	Depends on 6.2 (EV plug)	WARN
6.9	SetChargingProfile (update, 7.4 kW, stackLevel=1) → Accepted	Depends on 6.2 (EV plug)	WARN

Item	Test Description	Detail / Evidence	Result
6.10	MeterValues (Flow 1, after second SetChargingProfile)	Depends on 6.2 (EV plug)	WARN
6.11	StopTransaction (manual card tap, reason=Local)	Depends on 6.2 (EV plug)	WARN
6.12	StatusNotification Finishing (Flow 1, EV still plugged)	Depends on 6.2 (EV plug)	WARN
6.13	StatusNotification Available (Flow 1, EV unplug)	Depends on 6.2 (EV plug)	WARN
Flow 2: Remote Start with Charging Profile (6.14–6.26)			
6.14	StatusNotification Preparing (EV plug, Flow 2)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
6.15	RemoteStartTransaction → Accepted	HTTP 200	PASS
6.16	StartTransaction (triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
6.17	StatusNotification Charging (Flow 2)	Charging status not observed on MQTT in 20 s	WARN
6.18	MeterValues (Flow 2, before SetChargingProfile)	MeterValues not observed on MQTT in 15 s	WARN
6.19	SetChargingProfile (Flow 2, TxProfile 5 kW) → Accepted	HTTP 200	PASS
6.20	MeterValues (Flow 2, after first SetChargingProfile)	MeterValues not observed on MQTT in 15 s	WARN
6.21	SetChargingProfile (Flow 2, update 7.4 kW) → Accepted	HTTP 200	PASS
6.22	MeterValues (Flow 2, after second SetChargingProfile)	MeterValues not observed on MQTT in 15 s	WARN
6.23	RemoteStopTransaction → Accepted	HTTP 200	PASS
6.24	StopTransaction (triggered by RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN
6.25	StatusNotification Finishing (Flow 2, EV still plugged)	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing → go directly to Available</i>	WARN
6.26	StatusNotification Available (Flow 2, EV unplug)	Available not observed on MQTT in 30 s unplug EV	WARN

3 Notes on Failures and Warnings

SetChargingProfile command

The MEA sandbox REST API endpoint `POST /EV/remote/SetChargingProfile` accepts a `csChargingProfiles` object with `chargingProfilePurpose=TxProfile`, `chargingProfileKind=Absolute`, and a `chargingSchedule` containing one or more `chargingSchedulePeriod` entries with `chargingRateUnit=W`. Items 6.7, 6.9, 6.19, and 6.21 are PASS if the API returns HTTP 200.

EV-dependent items

- **6.2, 6.14** StatusNotification Preparing: require a physical EV to plug into the vSECC connector within the EV_WAIT_SEC timeout.
- **6.3–6.13** All Flow 1 session items depend on the EV being connected in item 6.2.

- **6.16–6.26** All Flow 2 session items depend on RemoteStart succeeding with an EV present (item 6.15 depends on EV presence from 6.14).

Item 6.11 — Manual stop

Item 6.11 tests that the EV driver can tap their RFID card on the vSECC to trigger a local `StopTransaction` with `reason=Local`. If no card tap is detected within `LOCAL_STOP_WAIT_SEC` seconds, the test falls back to RemoteStop to clear the session so Flow 2 can proceed. The fallback is recorded as **WARN** with a remark.

Items 6.6, 6.8, 6.10, 6.18, 6.20, 6.22 — MeterValues

The MEA sandbox REST API may return HTTP 404 for `TriggerMessage`. `MeterValues` are observable only when the vSECC autonomously publishes them to MQTT based on the `MeterValueSampleInterval` configuration. These items verify that `MeterValues` continue to be reported correctly both before and after each `SetChargingProfile` command.

Items 6.15, 6.23 — RemoteStart / RemoteStop

- **6.15** Uses a dedicated password (`MEA_PASS_START`) for the `/EV/cmd/chargepoint/remoteStart` endpoint.
- **6.23** RemoteStop with `transaction_id=0` succeeds when exactly one transaction is active; FAIL if no active transaction exists.

4 Dependency Map

Flow 1 6.2 → 6.3 → 6.4 → 6.5 → 6.6 → 6.7 → 6.8 → 6.9 → 6.10 → 6.11 → 6.12 → 6.13

Flow 2 6.14 → 6.15 → 6.16 → 6.17 → 6.18 → 6.19 → 6.20 → 6.21 → 6.22 → 6.23 → 6.24 → 6.25 → 6.26

Item 6.1 is independent of EV presence.